D110-D111 | Denominators and timing

PRIVATE SYNTHETIC teaching cases: economy indexes, separate annual retailer rupees, separate one-year lender crore. No current India data, actual store, bank, loan, investment or policy.

Attempt first	Work without notes before looking at references. Keep exact ratios, store demand drivers and lender repricing separate. No aggregate GDP elasticity or combined investment score. Actual confidence/errors/reviewer words/delayed transfer stay blank.
D110 | Exact economy ratios	Nominal GDP 200 -> 224, deflator 100 -> 106, population 100 -> 102. Real GDP 200 -> 211.320755, growth 5.660377%; real per capita growth 3.588605%, not subtraction 4%. Deflator is not the store basket CPI.
D110 | Independent store bridge	Customers 10000 -> 10200, frequency 4 -> 3.9, price 500 -> 520, unit cost 350 -> 370; new avoidable cost 400000. Baskets 40000 -> 39780, revenue 20000000 -> 20685600, after-cost contribution 6000000 -> 5567000. Required 42666.666667 expected or 42667 whole baskets, not proved demand.
D111 | Repricing timing	Assets 1000 at 10%, funding 800 at 6%. Funding 400 resets +200 bp immediately; loans 600 reset +150 bp after half-year. Baseline NII 52; first-year NII 48.5/NIM 4.85%, down 35 bp. Equal-year run-rate NII 53 is not first-year earned income.
D111 | Separate credit and contract	Credit expense 5 -> 12 plus opex 20 gives shocked pretax 16.5 versus baseline 27. Credit loss is not inside NII/NIM. Contract evidence reducing resetting loans 600 -> 300 gives NII 46.25/NIM 4.625%, pretax 14.25. No regulatory ECL or liquidity/economic-value inference.
Scope and stop	Static prose is Case-v1; revise after edits. Authored native arithmetic is not independent learner/source/full audiovisual/cross-viewer/account/legal clearance or overall 10/10. No expansion, lease, borrower/vendor contact, lending, deposits, repricing, derivatives, disclosure or spend authorized.

D110 macro inputs | One matched base

Two annual periods; nominal GDP in consistent arbitrary currency units, deflator on the SAME base 100, population in consistent units. No PPP/FX or base-year mixing.

Old nominal GDP	200.00
New nominal GDP	224.00
Old matched GDP deflator	100.00
New matched GDP deflator	106.00
Old population	100.00
New population	102.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D110 macro | Exact ratios, not demand

Deflator index divided by 100 converts nominal to real. Real per capita needs its own population denominator. Approximation error is shown in percentage points, not basis points.

Measure	Value
Old real GDP on deflator base 100	200.000000
New real GDP on same base	211.320755
Nominal GDP growth	12.0000%
Deflator growth	6.0000%
Population growth	2.0000%
Exact real GDP growth	5.6604%
Old real GDP per capita	2.000000
New real GDP per capita	2.071772
Exact real GDP per capita growth	3.5886%
Subtraction approximation, NOT exact	4.0000%
Approximation minus exact, percentage points	0.411395

GDP per capita is an average output measure, not each household income, disposable cash, inequality or a store forecast. A changed common deflator base preserves growth ratios but changes level units. Real data need release/revision date, matched base, population definition and permitted source evidence.

D110 retailer inputs | Build demand from units

Annual rupees; transactions per customer can be an expected average. New/downside use the SAME new unit economics and avoidable operating cost. Old unchanged fixed cost excluded from all paths.

Old customers	10000.00
Old annual transactions/customer	4.00
Old average rupees/basket	500.00
Old variable rupees/basket	350.00
New customers	10200.00
New annual transactions/customer	3.90
New average rupees/basket	520.00
New variable rupees/basket	370.00
New avoidable annual expansion operating cost	400000.00
Separate downside customers	9800.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D110 retail | Revenue can rise while contribution falls

Annual rupees, not crore. Thresholds require strictly positive unit margin; zero customer count makes required frequency undefined. Nonpositive margins do not authorize an expansion.

Measure	Old	New	Downside
Customers	10000.000000	10200.000000	9800.000000
Transactions/customer/year	4.000000	3.900000	3.900000
Baskets/year	40000.000000	39780.000000	38220.000000
Rupees/basket	500.000000	520.000000	520.000000
Variable rupees/basket	350.000000	370.000000	370.000000
Unit contribution, rupees/basket	150.000000	150.000000	150.000000
Revenue, rupees/year	20000000.000000	20685600.000000	19874400.000000
Variable expense, rupees/year	14000000.000000	14718600.000000	14141400.000000
Contribution before expansion cost	6000000.000000	5967000.000000	5733000.000000
Avoidable expansion cost	0.000000	400000.000000	400000.000000
Contribution after specified costs	6000000.000000	5567000.000000	5333000.000000
Change vs OLD after-cost contribution	0.000000	-433000.000000	-667000.000000
Required expected baskets to match OLD	40000.000000	42666.666667	42666.666667
Minimum whole baskets to match OLD	40000.000000	42667.000000	42667.000000
Required transactions/customer to match OLD	4.000000	4.183007	4.353741
Actual minus required expected baskets	0.000000	-2886.666667	-4446.666667

Baseline new revenue grows 3.428% but after-cost contribution falls 433000; downside falls 667000. Required volume is not available volume. No capex, tax, working-capital timing, financing or other fixed cost modeled, so not full profit, cash funding or NPV. Never revise macro GDP merely to force a store recommendation.

D111 lender inputs | Rate, balance and time

One year, crore, simple annual interest, constant balances and average earning assets. Delays are fractions of the year before one reset. No compounding, growth, repayments, fees, nonaccrual or tax.

Average earning assets / all loans, crore	1000.00
Old annual asset rate (fraction)	0.10
Funding balance, crore	800.00
Old annual funding rate (fraction)	0.06
Loans with contractual reset, crore	600.00
Asset reset delay, year fraction	0.50
Funding with contractual reset, crore	400.00
Funding reset delay, year fraction	0.00
Market shock, basis points	200.00
Asset pass-through of shock (fraction)	0.75
Funding pass-through of shock (fraction)	1.00
Annual operating expense, crore	20.00
Baseline annual credit loss expense, crore	5.00
Stress annual credit loss expense, crore	12.00
Contract-corrected reset loans, crore	300.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D111 repricing | Integrate each active time window

B/C are identical interest paths: C changes credit expense only. D changes only the contract-reset asset balance. First/second-half fractions are fractions of a full year, not local-half fractions.

Measure	Rate shock only	Rate + credit stress	Contract correction
Resetting asset balance	600.000000	600.000000	300.000000
Asset rate increment	1.5000%	1.5000%	1.5000%
Asset active fraction in first half	0.000000	0.000000	0.000000
Asset active fraction in second half	0.500000	0.500000	0.500000
Asset interest increment, first half	0.000000	0.000000	0.000000
Asset interest increment, second half	4.500000	4.500000	2.250000
Annual asset interest increment	4.500000	4.500000	2.250000
Funding rate increment	2.0000%	2.0000%	2.0000%
Funding interest increment, first half	4.000000	4.000000	4.000000
Funding interest increment, second half	4.000000	4.000000	4.000000
Annual funding interest increment	8.000000	8.000000	8.000000

Asset change 600*0.015*0.5=4.5; funding change 400*0.02*1=8. Contract correction asset change 300*0.015*0.5=2.25. All paths use average earning assets 1000, not funding 800. For arbitrary allowed delays, each half includes only its active portion.

D111 results | NIM is not credit, cash or economic value

All money crore; NIM uses average earning assets. Annual income is the sum of half-years, not annualized second-half income. Pretax excludes credit expense only AFTER NII.

Measure	Baseline	Rate shock	Plus credit stress	Contract corrected
First-half interest income	50.000000	50.000000	50.000000	50.000000
First-half funding expense	24.000000	28.000000	28.000000	28.000000
First-half net interest income	26.000000	22.000000	22.000000	22.000000
Second-half interest income	50.000000	54.500000	54.500000	52.250000
Second-half funding expense	24.000000	28.000000	28.000000	28.000000
Second-half net interest income	26.000000	26.500000	26.500000	24.250000
Annual interest income	100.000000	104.500000	104.500000	102.250000
Annual funding expense	48.000000	56.000000	56.000000	56.000000
Annual NII	52.000000	48.500000	48.500000	46.250000
NIM on average earning assets	5.2000%	4.8500%	4.8500%	4.6250%
NIM change vs baseline, bp	0.000000	-35.000000	-35.000000	-57.500000
Annual credit loss expense	5.000000	5.000000	12.000000	12.000000
Operating expense	20.000000	20.000000	20.000000	20.000000
Modeled pretax result	27.000000	23.500000	16.500000	14.250000
Pretax change vs baseline	0.000000	-3.500000	-10.500000	-12.750000
Fully repriced next-year NII, same balances	52.000000	53.000000	53.000000	48.500000
Naive annual balance gap x market shock	0.000000	4.000000	4.000000	-2.000000

Baseline NII 52/NIM 5.2%/pretax 27. Shock NII 48.5/NIM 4.85%, stress pretax 16.5; contract corrected NII 46.25/NIM 4.625%/pretax 14.25. Equal-year run rate 53 or 48.5 is conditional, not a forecast. Expense is not a cash charge-off, provision stock or regulatory ECL. No liquidity or economic value measured.

Audit | Separate cases, separate gates

Identities are authored arithmetic checks, not source assurance or real approvals. Broken downstream result references block their summaries, not independent cases.

Macro identity valid? (1=yes)	1.000000
Guarded exact real per capita growth	3.5886%
Retail identity valid? (1=yes)	1.000000
Guarded new after-cost contribution	5567000.000000
Lender identity valid? (1=yes)	1.000000
Guarded stressed pretax result	16.500000

GDP growth, store contribution and bank crore are not additive. Missing evidence/authority blocks real action even when the formula flag passes. Negative modeled contribution or NII remains visible, not suppressed as an invalid assumption.

Evidence | Known toy inputs, unknown real decisions

Original D110/D111 exercises at https://top1os.dev/. Existing selected OpenStax/FDIC teaching remains in the course; no new empirical source reading or full audiovisual review claimed.

Macro | Units and lineage	Macro inputs B6:B11 -> Macro -> Audit. Matched nominal/deflator/population periods and one base. Real source records need publisher, release/revision date, base/denominator and allowed audience; historical country examples are not current India facts.
Store | Units and lineage	Retail inputs -> Retail -> Audit. Customers x frequency x price/cost, annual rupees. Separate from GDP; no causal elasticity. Actual cohorts, footfall, conversion, capacity, avoidable cost, capex/cash timing still unknown.
Lender | Units and lineage	Lender inputs -> Repricing -> Lender results -> Audit. Static annual simple interest, crore; reset balances/timing/betas specified separately. Actual contracts, behavior, nonaccrual, liquidity, EVE and regulatory loss modeling absent.
Strongest authored objections	Aggregate average output may not reach store customers. Higher new revenue hides lower contribution. First-year repricing loss can differ from full-run-year margin. A changed contract reset balance moves affected asset income, not all rows via generic haircut.
Fictional day 0/30 review-stop	Macro/data role checks matched definitions; cohort/cash role tests store evidence; treasury/contracts/credit roles reconcile reset terms and credit expense. No real people accepted these roles. Stop real expansion, financial changes or disclosures without actual evidence and authority.
Fresh delayed context	Without notes change deflator/population, customer frequency, then lender delay to one quarter and pass-through. Rebuild the exact ratios, required volume and time weights. Record actual new response/error/correction/objection/result only if performed; no invented mastery or accepted approval.

Actual work | No performed learner or reviewer evidence

Attempt, confidence, source, actual error, correction, objection and delayed novel context remain blank because no learner/reviewer execution is supplied.

Actual date / case	Attempt / confidence	Exact input / source	Observed error / claim	Correction / decision	Actual objection / response	Delayed context / result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline exact ratios / store / lender	PASS: all baseline formulas independently recalculated; three unit-separated cases.
Macro alternate base / population	PASS: common deflator-base change preserves growth; new population and zero nominal recompute.
Retail price / volume / margin cases	PASS: changed frequency/price/cost/customers and zero/negative margins, baseline restored.
Loan reset timing / funding delay	PASS: quarter/immediate asset reset and half-year funding reset use correct active windows.
Pass-through / shock / contract correction	PASS: half pass-through, zero/negative shock and corrected contract balance recompute.
No reset / full year / quarter year	PASS: full-year no reset, zero funding reset and quarter-year cases checked.
Invalid inputs / case isolation	PASS: 15 invalid-input cases block affected model, independent models preserved.
Broken result references / isolation	PASS: three broken result references block affected audit, independent cases preserved.
Restored baseline / actual blanks	PASS: baseline restored, saved formula caches error-free, actual fields blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.